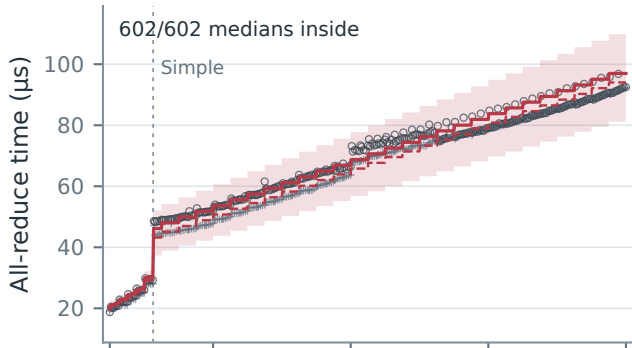


# NCCL protocol work explains the timing step

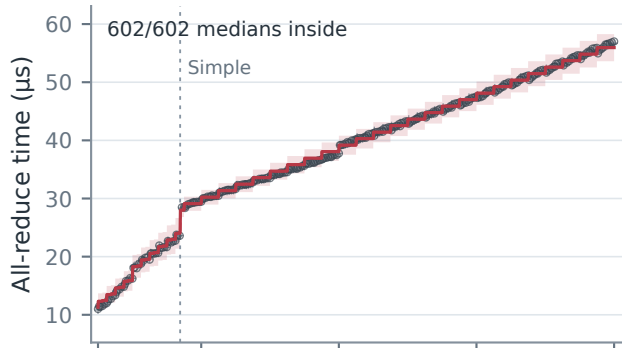
Development comparison | 301 payloads per curve | Both methods measured on Merlin

— Model: GPU events    Software + method band    + Benchmark measurement  
- - - Model: benchmark    ○ GPU-event measurement    × Outside band

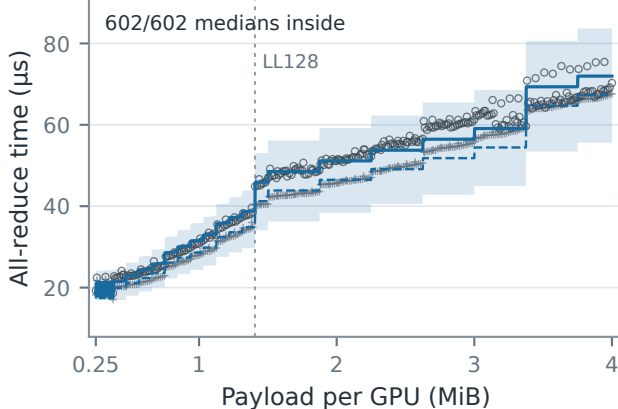
A100 | 2 GPUs



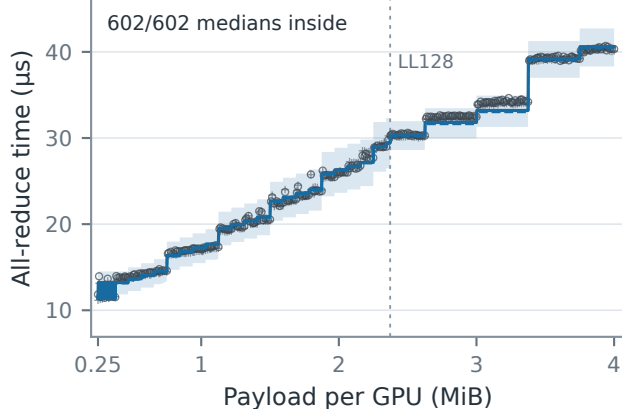
GH200 | 2 GPUs



A100 | 4 GPUs



GH200 | 4 GPUs



Band: calibration residuals, repeat spread and method allowance. Gold crosses mark uncovered medians.

LL carries data and flags together; Simple fences data, publishes counters, and polls in the same kernel.